

THAMINDU SAMARATHUNGA

Computer Science Undergraduate | UI/UX Developer

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PROFILE

Final-year BSc (Hons) Computer Science student at NSBM Green University (affiliated with the University of Plymouth), with hands-on experience building full-stack AI-powered platforms and IoT systems. Proficient in React, TypeScript, Python, and FastAPI, with a strong foundation in machine learning and data-driven product development. Actively seeking software engineering intern or graduate roles where I can contribute to impactful projects.

TECHNICAL SKILLS

Build with daily: Python, JavaScript, TypeScript, React, FastAPI, SQL/SQLite, Git & GitHub

Also know: C#, C, HTML/CSS, PyTorch, scikit-learn, Computer Vision (VITON-HD, MediaPipe, SegFormer), MERN Stack, Ethereum Smart Contracts (Solidity), Flutter, Firebase, IoT (ESP32)

Learning right now: Laravel & PHP, Next.js, Node.js

LEADERSHIP & ACTIVITIES

Top Board Member & Event Organizer — NSBM Chess Club

Dec 2023 – Dec 2024

- Earned a seat on the 2024 club Top Board after a year of running events that members actually showed up to.
- Helped organise NSBM USports 2023, one of Sri Lanka's biggest inter-university sports festivals, as an organising-committee member across a multi-week run.

Industry Preparation Training Programme 2025 (IPTP)

Jul – Aug 2025

Faculty of Computing, NSBM Green University — two-week intensive on professional development practices

Member — NSBM Music Club

EDUCATION

BSc (Hons) Computer Science

Mar 2023 – Nov 2026 (expected)

University of Plymouth (delivered at NSBM Green University, Sri Lanka)

Relevant coursework: Software Development, Data Structures & Algorithms, Statistics, Front-End Development, AI/ML, HCI.

G.C.E. Advanced Level — Physical Science Stream

2021

R/St. Aloysius College

CERTIFICATIONS

- Microsoft Learn — Using GitHub Copilot with Python, Introduction to GitHub, Introduction to Python
- INFINITY'23 — Mathematics & Statistics Circle, NSBM Green University.

PROJECTS

TOP LABEL — AI Virtual Try-On & Fashion Recommendation Platform

Oct 2025 – May 2026

Final-Year Project · University of Plymouth / NSBM Green University · React · FastAPI · Python · VITON-HD · SQLite

- Hit 96% accuracy on body-type matching by training and comparing two models — Random Forest (96%) vs HistGradientBoosting (93%) — on 100,000 synthetic South Asian body measurement samples, then layering a KNN cosine-similarity recommender on top.
- Built a 9-stage computer-vision pipeline (MediaPipe pose detection → SegFormer segmentation → GrabCut masking → VITON-HD) to turn a single uploaded photo into a realistic 768×1024 try-on result.
- Shipped end-to-end as a working product: React/TypeScript storefront, Python FastAPI backend, SQLite, and an admin dashboard for products, users, and live stats — all wired together, not just slides.

Industry-Based Big Data Analytics Using Apache Spark

Jan 2026 – Apr 2026

Academic Project · University of Plymouth · Apache Spark · Python · Databricks · ML

- Built a big data analytics pipeline on the Olist Brazilian E-Commerce dataset (100,000+ orders) using Apache Spark and Spark SQL to uncover business insights on revenue, delivery performance, and customer satisfaction.
- Produced four business visualisations in Databricks using Spark SQL and Matplotlib: Monthly Revenue Trend, Order Status Distribution, Delivery Status vs Average Review Score, and Top 10 States by Revenue.
- Trained a Random Forest ML model that achieved 82.6% accuracy in predicting low customer review outcomes, proving delivery reliability is the #1 driver of satisfaction.

FurnitureViz — 3D Room Design & Furniture Visualization

Jan 2026 – Mar 2026

Academic Project · PUSL3122 HCI · Next.js · Three.js · React · Firebase

- Implemented real-time 3D rendering using Three.js and WebGL, converting 2D furniture coordinates into interactive 3D space with XZ plane mapping, scale controls, and realistic scene lighting.
- Integrated 100+ furniture models across multiple categories with directional and ambient lighting for photorealistic scene rendering.
- Built with Next.js 16, TypeScript, React Three Fiber, Three.js, Firebase Firestore, Zustand state management, and Tailwind CSS for a polished, responsive user experience.

Ceylon WorkForge — Micro Job Platform for Sri Lanka

Jan 2025 – May 2025

Academic Project · University of Plymouth · Next.js · Node.js · Firebase · Tailwind

- Developed a full-stack micro job marketplace connecting freelancers, job seekers, employers, and small businesses through a centralized digital platform for Sri Lanka's growing gig economy.
- Owned the Admin Dashboard: implemented user management, job seeker profiles, featured listings, job board management, report handling, and payment workflow integration using Next.js and TypeScript.
- Built with Next.js, Node.js, Firebase Authentication, Firestore Database, and Tailwind CSS. Integrated key features like real-time messaging, issue reporting, and admin controls.

Vehicle Parts Marketplace Mobile App

Feb 2025 – May 2025

Academic Project · University of Plymouth · Flutter · Mobile Development

- Built a mobile marketplace app that simplifies buying and selling of vehicle spare parts by providing a centralized platform where users can securely post ads, search by vehicle type/brand/location, and connect with sellers.
- Implemented user authentication (login & registration), advanced search and filtering (vehicle type, brand, location, price), categorized listings (engines, wheels, brakes, accessories), and direct seller communication.
- Focused on usability, transparency, and real-world problem-solving — addressing fragmented marketplaces and inefficient searching in the vehicle parts industry.

Flood Sense — IoT Flood Disaster Management System

Jan 2025 – May 2025

Academic Project · ESP32 · Firebase · Flutter · IoT · Team Collaboration

- Designed and built an ESP32-based sensor network for real-time flood monitoring in Sri Lanka, streaming water-level, temperature, humidity, pH, soil moisture, and rainfall data to Firebase.
- Developed a Flutter mobile app delivering live sensor updates, push notifications for threshold breaches, emergency contact management, and water-level architecture visualizations.
- System powered by solar-charged battery with GSM module for reliable remote communication, designed for scalability and ease of deployment in flood-prone regions.